Impact of Datasets on Generative Models in Imaging

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Abstract

Quality and diversity assessments of generated image datasets are vital to choose and improve algorithmic generators. Based on summary statistics we present a framework to compute established generative model performance indices like FID, Precision, and Recall on different feature layers of different deep convolutional neural networks. Building on this framework we model neural network activation distributions with Extreme Value Theory (EVT). We observe that looking at layers deeper within neural networks the clustering behavior of maximum values of activation distributions becomes more distinct. Based on EVT we propose the Extreme Value Index to Compare Image Distributions (EVI). The proposed index is supposed to measure mode coverage, generated artifacts and quality of a generated dataset. However, in a series of empirical tests the index fails to definitely distinguish between declines of quality or diversity of generated datasets.

Zusammenfassung

Die Bewertung von Qualität und Diversität generierter Bilder ist von entscheidender Bedeutung bei der Wahl von algorithmischen Bild-Generatoren und deren Verbesserung. Basierend auf deskriptiven Statistiken präsentieren wir ein Framework zur Berechnung etablierter Leistungs-Indexe generativer Modelle wie FID, Precision und Recall auf verschiedenen Feature-Layern unterschiedlicher Deep Convolutional Neural Networks. Aufbauend auf diesem Framework modellieren wir Aktivierungsverteilungen neuronaler Netze mit Extreme Value Theory. In tieferen Feature Layern beobachten wir ein eindeutigeres Cluster verhalten der Aktivierungen. Außerdem stellen wir den Extreme Value Index to Compare Image Distributions vor. Der vorgeschlagene Index soll messen, wie divers ein Datenset ist, ob der Generator Artefakte erzeugt und wie gut die Qualität der generierten Bilder ist. In einer Reihe von Tests kann der vorgeschlagene Index jedoch nicht zwischen Qualitätsverlusten oder abnehmender Vielfalt eindeutig unterscheiden.

Contents

1 Introduction

In recent years generative image modeling has been researched extensively. Many different approaches are being pursued, for example Generative Adversarial Networks (GANs), ([Goo+14]), variational autoencoders ([KW22]), and diffusion models ([Soh+15]). While no single approach has distinguished itself to succeed over the others some models are performing better on specific tasks than others ([Cod23]). For example, stable diffusion is a popular free-to-use text-to-image diffusion model capable of generating photo-realistic images given any text input ([Wik23]). In a scientific context, GANs were for example deployed to model high-resolution solar images ([Che+23]) and to make temporal predictions about crop growth ([Dre+21]). Especially for scientific purposes, researchers need to know about shortcomings of their models, i.e. does the model accurately generate natural images and does it produce a wide variety of images such that they can be used for simulations. For instance, even though [Che+23] achieved generating solar images that experts could not distinguish from real ones with a GAN, they are assuming that this model fails to generate a sufficiently diverse sample of images for their objective. Conventional GAN performance indices, like the Fréchet Inception Distance (FID) ([Heu+18]), or Precision and Recall ([Saj+18]), indicate that the generative model generates images that approximate the real image distribution well. However, the low sample variety was detected when real samples could not be adequately approximated by the model. While currently a task specific validation of the approximation capabilities of a generative model might be indispensable, in this work we focus on widely applicable generative image model performance indices that give a first impression about the model performance. In the next section 1.1, we introduce generative image models by the example of GANs. Next in 1.2, we review some established image model performance indices (MPI) based on activations inside deep neural networks. In 2, we derive from a statistical point of view and based on extreme value theory (EVT) a new MPI, the Extreme Value Indicator to Compare Image Distributions (EVI). In section 3, we discuss the implementation of the EVI. We test and compare the EVI with the MPIs presented in section 1.2 in 4.

Contributions:

- We introduce a framework to model the distributions of activations inside deep convolutional neural networks founded on extreme value theory.
- Based on this framework we present a new MPI that distinguishes the capability of a generator to cover the whole sample space of a real dataset, generate high perceptual quality images, and whether generated images exhibit artifacts.

- Further, we compute and compare established MPIs on different feature layers of different convolutional neural networks.

1.1 Generative Adversarial Networks

Usually, generative imaging models are deep neural networks that learn to approximate a real distribution of images. To achieve this objective, they are usually trained on unlabeled images. Since [Goo+14] proposed the first GAN architecture many advances improved the architecture (i.e. [KLA19], [BDS19], [SSG22]) but the original building blocks have not changed.

On the one side, we have the generator G_{θ_g} that samples random numbers from a predefined distribution and generates an image that should look like the images from the real sample. G_{θ_g} usually consists of several convolutional building blocks which weights we denote by θ_g. Working antagonistically against the generator there is the discriminator D_{θ_d}, which normally is based on a deep convolutional neural classifier network, which parameters we denote by θ_d. The discriminator receives generated images from the generator and real images with the objective to distinguish between them. The two players try to optimize a minimax game that can be defined in respect to the following value function:

$$\begin{aligned}&\min_{\theta_g}\max_{\theta_d} V(G_{\theta_g}, D_{\theta_d}) = \\ &E_{\boldsymbol{x}\sim p_{data}(\boldsymbol{x})}[\log D_{\theta_d}(\boldsymbol{x})] + E_{\boldsymbol{z}\sim p_z(\boldsymbol{z})}[\log(1 - D_{\theta_d}(G_{\theta_g}(\boldsymbol{z}))]\end{aligned} \tag{1.1}$$

where p_{data} denotes the real image distribution and p_z the predefined distribution from which the generator samples. From here the role of MPIs follows. They should give us information about how well our generative model can produce images that could belong to the training dataset and how diverse the generated images are. More specifically, whether for $\boldsymbol{z} \sim p_z(\boldsymbol{z})$ $G_{\theta_g}(\boldsymbol{z}) \sim p_{data}$. Due to their success we focus on MPIs based on the feature layers inside neural networks.

1.2 Feature Based MPIs

[Sal+16] were the first to use the output of an ImageNet ([Den+09]) pretrained Inception-V3 network through which a sample of generated images was fed ([Sze+15]) to calculate the so-called Inception Score to measure the fidelity and diversity of generated images. They argue that if a generator creates images of good quality the inception network should assign them to one label with high probability. If the generator is creating a diverse sample of images the assigned labels should be approximately uniformly distributed. Combining these two arguments they defined the Inception Score:

$$IS = \exp\left[E_{x\sim p_g}\left(D_{KL}\left\{p_d(\cdot|x)||E_{x\sim p_g}[p_d(\cdot|x)]\right\}\right)\right] \tag{1.2}$$

where p_g is the distribution of the generated images, $p_d(y|x)$ is the probability that the discriminator ascribes to the image x having the label y, and D_{KL} the Kullback-Leibler-Divergence. Clearly, this score depends rather heavily on the ImageNet classes and the classification capability of the pretrained network and therefore might only be reasonable for ImageNet like datasets.

However, their approach was taken on by [Heu+18] to develop the Fréchet Inception Distance (FID) but this time the penultimate layer activations were used to calculate a distance measure, the Fréchet Distance, between real and generated images. The key idea behind this approach is to embed real and generated images into a vision-relevant feature space (see Figure 1) and compute a distance between the two embedded distributions. These distributions are assumed to be normally distributed, and therefore the distance can be quantified using the Fréchet Distance ([DL82]):

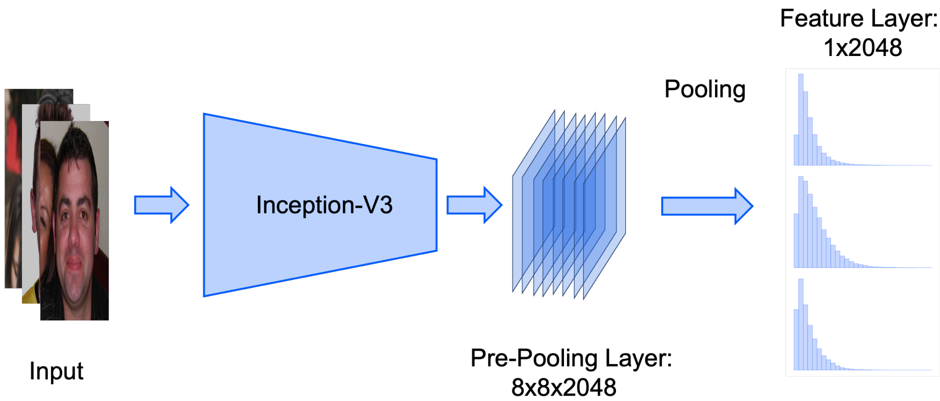


Figure 1: Overview of how images are embedded into the penultimate feature-layer of the InceptionNet-V3. On the right of the graphic special attention is payed to single feature distributions.

$$FID(\mu_r, \Sigma_r, \mu_g, \Sigma_g) = ||\mu_r - \mu_g||_2^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}}) \tag{1.3}$$

where (μ_r, Σ_r), and (μ_g, Σ_g) denote the sample mean and covariance of real and generated data embeddings, respectively, and $Tr(\cdot)$ is the matrix trace. FID has been found to correlate reasonably well with human judgments of generated image fidelity ([Heu+18]; [Xu+18]; [Luc+18]). A drawback of the FID (and the Inception Score) is that it measures both image quality and diversity with a single score. Hence, a high FID can mean either low diversity, poor quality, or both.

[Saj+18] tackled that problem and introduced the Precision and Recall indicator which they argue measure the quality and diversity of a generated sample, respectively. They also use the activation distributions of a sample of generated images P_G and real images P_R of the penultimate layer of the

InceptionNet and define Precision and Recall as the set of all $\alpha, \beta \in (0, 1]$ for which exist μ, ν_R, ν_G such that

$$P_G = \beta\mu + (1 - \beta)\nu_R \quad and \quad Q = \alpha\mu + (1 - \alpha)\nu_G. \tag{1.4}$$

The component ν_R corresponds to the region of P_R where the generator fails to create any images and ν_G to images that the generator creates but are not in the real distribution of images. μ corresponds to the region where both real images and generated images lie. However, as we will see in section 4.4 the Precision and Recall score fail to distinguish between quality or diversity of certain generated datasets.

All feature based MPIs assume that the features have general perceptual relevance, that is the features are relatively independent of the feature network training data and an improvement of an MPI in the feature space corresponds to perceptual higher quality images. [Kyn+22] show that this assumption might not be met by the penultimate feature layer of the InceptionNet since the feature correspond strongly to the ImageNet classes. Thus, they are not independent of the training data. Therefore, when there is a domain gap between the ImageNet classes and the generative model training data the MPIs based on InceptionNet features of the penultimate layer might become unreliable. Nonetheless, comparing image distributions based on their feature activations is a promising approach. [Zha+18] showed that distance measures computed on feature maps of neural networks correspond more with human perceptual judgment than comparing images in their original domain. To take advantage of the effectiveness of feature maps but not be limited to a single layer we introduce a framework to compute a MPI that can be computed on different feature layers in different networks.

2 Extreme Value Index to Compare Distributions

A classifier's task is to map an image to its true class. We hypothesize that an image belonging to class A should score high on features intrinsic to class A while scoring low on all other features. If we assume that our classifier can distinguish images from many different domains images might score low on most feature combinations so that the classifier can distinguish between the classes. This scoring behavior is enforced by activation functions like the maximum or the ReLU-function (rectified linear unit activation function). Thus, a classifier is incentivized to choose its weights such that signals important for classification achieve high scores since otherwise, they might fail to pass activation functions. From this line of reasoning, in favor to decode the important information inside a neural network it might make sense to model the distribution of maximum values inside a neural network. In this section, we describe the problem which MPIs want to solve from a statistical point of view and base the arguments from above on a mathematical foundation. Since we are venturing out from a statistical point of view we first define the objects we are considering:

Definition 2.1. *Let $\mathcal{M} = \{\mathcal{A}, \mathcal{F}, \mathbb{P}_G, \mathbb{P}_G\}$ be the statistical model where $\mathcal{A}$ is the activation sample space $\mathbb{R}^d$ and $\mathcal{F}$ the respective σ-algebra. $\mathbb{P}_G$, and $\mathbb{P}_R$ are the probability measure for activations of generated images and the real images, respectively.*

For instance, when we consider the pre-pooling layer from Figure 1 we think of $\mathcal{A}$ as the activation space $\mathbb{R}^{2048}$ with 64 observations.

For simplicity, we assume that these observations are independent and identically distributed (*iid*) even though the observations are spatially related and therefore this assumption is hardly justified. Now, we pose the problem:

Problem 2.2. *We want to find a statistic that indicates how well $\mathbb{P}_G$ approximates $\mathbb{P}_R$.*

A common approach to such a problem is to look at the likelihoods of the distributions $\mathbb{P}_G$ and $\mathbb{P}_R$ and develop a statistical test to see whether the two distributions differ significantly from each other. However, approximating the likelihoods and evaluating them for a given image dataset poses a difficult task and it has been shown that likelihood based approaches often cannot differ between high or bad quality images ([TOB16], [Wu+17]).

Therefore, in the first step we want to make a simple summary of the model defined in 2.1 to facilitate statistical interference over the image samples. For this purpose, we look at the concept of summary statistics described in [CB21]. Summary statistics should accomplish to summarize a set of observations to preserve the largest amount of information as succinctly as possible. In the next section, we specify this intuitive definition of summary statistic.

2.1 Summary Statistics

We consider the observations $\boldsymbol{A_1}, \ldots, \boldsymbol{A_n}$, each a random vector in $\mathcal{A}$. Since we assume the observations are identically distributed we might characterize the sample through their distributions. We first focus on the one dimensional samples $X_j = A_{i,j}$ with $j = 1, ...n$ over $i = 1, ...d$ where $A_{i,j}$ is the projection of $\boldsymbol{A_j}$ on the ith coordinate. For simplicity, we assume the distributions of X_j are absolutely continuous and parameterized by a parameter $\theta \in \mathbb{R}$. Hence, we have a probability density function (pdf) ρ dependent on one parameter. Moreover, since we assume that the observations are independent we can consider a product model for the random vector $\boldsymbol{X} = (X_1, ..., X_n)$. We use the definition of a sufficient statistic from [CB21] to specify what preserving a large amount of information means.

Definition 2.3 (Sufficient Statistic.). *A statistic $T(\boldsymbol{X})$ is a sufficient statistic for θ if the sample conditional distribution $f(\boldsymbol{X}|T(\boldsymbol{X}))$ does not depend on θ.*

In other words $T(\boldsymbol{X})$ represents the sample distribution in place of θ, and thus we could refer from $T(\boldsymbol{X})$ back to the properties of our sample. However, a sufficient statistic does not necessarily reduce the data at hand. For instance, the entire ordered sample $T(\boldsymbol{X}) = (X_{1:n}, ...X_{n:n})$ is a sufficient statistic.

Therefore, [CB21] introduce minimal sufficient statistics (MSS):

Definition 2.4 (Minimal Sufficiency). *A sufficient statistic $T(\boldsymbol{X})$ is called a minimal sufficient statistic if for any other sufficient statistic $T'(\boldsymbol{X})$, $T(\boldsymbol{X})$ is a function of $T'(\boldsymbol{X})$, i.e., $T(\boldsymbol{X}) = f(T'(\boldsymbol{X}))$.*

Hence, we could say that an MSS contains information about the sample more succinctly than any other sufficient statistic. For our purposes, an MSS thus fulfils the requirements of a summary statistic to reduce the size of the available data while maintaining the information. The next question that presents itself is how to find an MSS for the sample $\boldsymbol{X}$.

2.2 Maximum Statistic

Since we consider the activation distribution of a classifier neural network and a classifier should distinguish between different distributions of classes, ideally the feature activations should be easily separable. For large k parameter the Weibull distribution might be appropriate to model such distributions since it decreases rapidly to both sides of its peak. Hence, two Weibull distributions are easily separable. We show that the maximum function approximates an MSS for the Weibull distribution for large k-parameter:

Proposition 2.5.

1. *Let* $\boldsymbol{X} = (X_1, ..., X_n)$ *be iid Weibull distributed random variables with pdf* $\rho(x) = k\lambda(\lambda x)^{k-1}e^{-(x\lambda)^k}\mathbb{1}_{\{x>0\}}(x)$, $k > 0$.
   Then $T(\boldsymbol{X}) = (\sum_{i=1}^n X_i^k)^{1/k}$ *is an MSS for* λ.
2. *Let* $Y_k = (\sum_{i=1}^n X_i^k)^{1/k}$. *Then* $Y_k \xrightarrow[k\to\infty]{} \max\{X_1, ..., X_n\}$ *almost surely.*

Proof. Let $\rho^{\otimes n}$ be the joint pdf of ρ. From [CB21] Theorem 6.2.13 it follows that $T(\boldsymbol{X})$ is an MSS if for all $\boldsymbol{x} = (x_1, ..., x_n)$, and $\boldsymbol{y} = (y_1, ..., y_n)$ $\rho^{\otimes n}(\boldsymbol{x})/\rho^{\otimes n}(\boldsymbol{x})$ is constant as a function of λ if and only if $T(\boldsymbol{x}) = T(\boldsymbol{y})$. Hence, we need only to show that the quotient $\tilde{\rho}^{\otimes n}(\boldsymbol{x})/\tilde{\rho}^{\otimes n}(\boldsymbol{y})$ is a constant function of λ with $\tilde{\rho}^{\otimes n}(x) = \lambda^{nk}e^{-\lambda^k \sum_{i=1}^n x_i^k}$. We see that

$$\frac{\tilde{\rho}^{\otimes n}(\boldsymbol{x})}{\tilde{\rho}^{\otimes n}(\boldsymbol{y})} = e^{-\lambda^k(\sum_{i=1}^n y_i^k - \sum_{i=1}^n x_i^k)} \tag{2.1}$$

is a constant function if and only if $T(\boldsymbol{x}) = T(\boldsymbol{y})$.

Next, we show the second statement. Since all sets for which at least two X_i are equal are Lebesgue nul sets we can assume without loss of generality that $X_1 > X_i$ for $i = 2, ...n$. Then

$$\lim_{k\to\infty}\left(\sum_{i=1}^n X_i^k\right)^{1/k} = \lim_{k\to\infty}(X_1)\left(1 + \sum_{i=2}^n\left(\frac{X_i}{X_1}\right)^k\right)^{1/k} = X_1. \tag{2.2}$$

Since $X_1 > X_i$, $X_i/X_1 < 1$, hence $1 \leq 1 + \sum_{i=2}^n \left(\frac{X_i}{X_1}\right)^k < \infty$ for all k. Thus, $\left(1 + \sum_{i=2}^n\left(\frac{X_i}{X_1}\right)^k\right)^{1/k}$ converges to 1 for $k \to \infty$. In particular, $Y_k \xrightarrow[k\to\infty]{} \max\{X_1, ..., X_n\}$ everywhere but on a Lebesgue nul set. $\square$

However, if the Weibull distribution does not fit the activation distribution the maximum statistic might still preserve enough information about the sample. When we assume nothing about the distribution of the activations, [Jay57] recommend using a maximum entropy distribution. In the case of bounded distributions, the uniform distribution is a max entropy distribution ([CT05]). Further, it turns out that the maximum is a sufficient statistic for a uniform distribution.

Proposition 2.6. *Let* $(X_1, ..., X_n)$ *be iid uniform distributed random variables with pdf* $\rho(x) = \theta^{-1}\mathbb{1}_{x\in(0,\theta)}(x)$. *Then* $T(\boldsymbol{X}) = \max\{X_1, ..., X_n\}$ *is a sufficient statistic for* θ.

Proof. Analog to [CB21] example 6.2.8 we can use the Fisher–Neyman factorization theorem which states that T is sufficient for θ if and only if non-negative functions g and h can be found such that $\rho^{\otimes n}(\boldsymbol{x}) = g(T(\boldsymbol{x}))h(\boldsymbol{x})$

where only g depends on θ for all $\boldsymbol{x}$ in the sample space. We just have to switch the probability mass function for which [CB21] show proposition 2.6 in Example 6.2.8 for a pdf:

$$\begin{aligned}\rho^{\otimes n}(\boldsymbol{x}) &= \theta^{-n}\prod_{i=1}^{n}\mathbb{1}_{\{x_i\in(0,\theta)\}}(x_i)\\ &= \theta^{-n}\mathbb{1}_{\{\boldsymbol{x}\in(\boldsymbol{0},\boldsymbol{\theta})\}}(\max\{x_1,...,x_n\})\prod_{i=1}^{n}\mathbb{1}_{x_i\in(0,\infty)}(x_i).\end{aligned}\tag{2.3}$$

The statement now follows from the Fisher–Neyman factorization theorem with $g(T(\boldsymbol{x})) = \theta^{-n}\mathbb{1}_{\{\boldsymbol{x}\in(\boldsymbol{0},\boldsymbol{\theta})\}}(\max\{x_1,...,x_n\})$ and $h(\boldsymbol{x}) = \prod_{i=1}^{n}\mathbb{1}_{x_i\in(0,\infty)}(x_i)$. □

Remark 2.7. While the maximum is a sufficient statistic for the uniform distribution it is not a minimal sufficient statistic as follows directly from [CB21] Theorem 6.2.13 used to prove proposition 2.2.

Since we model layers of neural networks specifically after ReLU functions it justifies assuming the lower bound 0 in proposition 2.6.

An additional advantage of the maximum statistic in contrast to other established summary statistics such as the mean or the variance is that it is undistored by non-linearities such as the ReLU or the maximum function which are fundamental elements of neural network architectures.

Now, we can considerably reduce the data at hand: After applying the maximum function over the observations per feature we obtain the random vector

$$\boldsymbol{S} = (S_1,...,S_d) = (\max\{A_{1,1},...A_{1,n}\},...,\max\{A_{1,d},...A_{d,n}\}).$$

To model the distribution of the random vector $\boldsymbol{S}$ we are going to leverage extreme value theory.

2.3 Extreme Value Theory

Usually, Extreme Value Theory (EVT) is applied to model extreme hazards to estimate the risk that a particular communal or industrial damage occurs. For example in hydrology, EVT is used to estimate the probability of floodings ([BT98]) or in financial risk management to forecast stock market shocks ([EKM13]). A standard example is as follows ([HF06]): A government wants to protect its shores against flooding. Therefore, it builds dikes to prevent flooding. Since these dikes are expensive to build the government determines the dikes should be built sufficiently high that the probability of a flood (i.e. the seawater level exceeding the top of the dike) in a given year is below some small probability. To estimate the height of the dikes only the yearly maximum sea levels are considered and their distribution is modeled with EVT.

Analog to [HF06], we start with establishing the univariate theory of extreme values from which the multidimensional case follows easily. Let $(X_1, ..., X_n)$ be *iid* random variables with distribution function $F(x)$. If we naively take the maximum value of all random variables we will obtain a degenerate distribution

$$P(\max\{X_1, ..., X_n\} \leq x) = P(X_1 \leq x, ..., X_n \leq x) = F^n(x) \tag{2.4}$$

which converges to 0 if $x < x^*$ and to 1 if $x \geq x^*$ with $x^* = \sup\{x : F(x) \leq 1\}$. To obtain a non-trivial distribution, a normalization is necessary.

Definition 2.8. *Suppose there exist sequences of constants $a_n \geq 0$, and $b_n \in \mathbb{R}$, $n = 0, 1, ...$ such that*

$$\frac{\max\{X_1, ..., X_n\} - b_n}{a_n} \tag{2.5}$$

has a non-degenerate limit distribution $G(x)$,

$$\lim_{n\to\infty} F^n(a_n x + b_n) = G(x). \tag{2.6}$$

Then we say F lies in the domain of attraction of G.

The possible limit distributions are called extreme value distributions (EVD) and have all the following form (for a detailed proof of the following theorem see [HF06] theorem 1.1.3):

Theorem 2.9 (Fisher–Tippett–Gnedenko Theorem)**.** *The class of extreme value distributions is $G_\gamma(ax + b)$ with $a \geq 0$, $b \in \mathbb{R}$, where*

$$G_\gamma(x) = \exp\left[-(1 + \gamma x)^{-\frac{1}{\gamma}}\right], \quad 1 + \gamma x \geq 0, \tag{2.7}$$

with γ real and where for $\gamma = 0$ the right-hand side is interpreted as $\exp(-e^{-x})$.

Next, we have to answer the question of which distributions satisfy definition (2.8) or rather specifically, whether it holds for the Uniform or the Weibull distribution which we use to model the activation distributions.

Proposition 2.10. *Let X be a Weibull distributed random variable with pdf $\rho(x) = k\lambda(x\lambda)^{k-1}e^{-(x\lambda)^k}\mathbb{1}_{\{x>0\}}(x)$, $k > 0$. Then X is in the domain of attraction of an EVD G with $\gamma = 0$.*

Proof. The distribution function of X is $F(x) = 1 - e^{-(\lambda x)^k}$. Then from

[HF06] theorem 1.1.8 it follows we have to show that $\lim_{t\to\infty}\left(\frac{1-F}{F'}\right)'(t) = 0$.

$$\begin{aligned}\lim_{t\to\infty}\left(\frac{1-F}{F'}\right)'(t) &= \lim_{t\to\infty}\left(\frac{e^{-(x\lambda)^k}}{k\lambda(x\lambda)^{k-1}e^{-(x\lambda)^k}}\right)'(t)\\ &= \lim_{t\to\infty} -(k-1)k(t\lambda)^{-k}\\ &= 0.\end{aligned} \tag{2.8}$$

□

Proposition 2.11. *Let X be a uniform distributed random variable with pdf $\rho(x) = \frac{1}{\theta}\mathbb{1}_{x\in(0,\theta)}(x)$. Then X lies in the domain of attraction of an EVD with $\gamma = -1$.*

Proof. Let $F(x)$ be distribution function of X. Then from [HF06] corollary 1.1.13 we have to show that $\lim_{t\to\theta}\frac{(\theta-t)F'(t)}{1-F(t)} = 1$.

$$\lim_{t\to\theta}\frac{(\theta-t)F'(t)}{1-F(t)} = \lim_{t\to\theta}\frac{(\theta-t)(\theta)^{-1}}{1-(\theta)^{-1}t} = 1. \tag{2.9}$$

□

Next, we look at how we might model the dependence structure of the random vector $\boldsymbol{X} = (X_1, ..., X_d)$. Analog to [Bei+04] we introduce the multivariate extreme value distributions (MEVD). For this purpose, we first introduce the notion of convergence of random vectors.

Definition 2.12. *Let $d \in \mathbb{N}, (\boldsymbol{Y_n})_{n\geq 1}$ be a sequence of $\mathbb{R}^d$-dimensional random vectors and $\boldsymbol{Y}$ be a random vector with distribution function Q. Then $\boldsymbol{Y_n}$ converges in distribution to $\boldsymbol{Y}$ if for all Borell-Sets A with $Q(\partial A) = 0$ $\mathbb{P}(Y_n \in A) \xrightarrow{D} Q(A)$ for $n \to \infty$. Where ∂A is the topological boundary of A.*

As in the last section, we assume that there exist sequences of vectors $\boldsymbol{a_n} > 0$, $\boldsymbol{b_n} \in \mathbb{R}^d$ such that the random vector $\max\{\boldsymbol{X_1}, ..., \boldsymbol{X_n}\}$ converges in distribution after standardization:

$$\lim_{n\to\infty}\mathbb{P}\left(\frac{\max\{\boldsymbol{X_1}, ..., \boldsymbol{X_n}\} - \boldsymbol{b_n}}{\boldsymbol{a_n}} \leq \boldsymbol{x}\right) = \lim_{n\to\infty} F^n(\boldsymbol{a_n}\boldsymbol{x} + \boldsymbol{b_n}) = G(\boldsymbol{x}). \tag{2.10}$$

For $j = 1, ...d$ let F_j and G_j be the marginals of the distribution functions F and G, respectively. For instance, $F_1(x) = F(x, \infty, ..., \infty)$. By assumption our marginals are not degenerated and since a sequence of random vectors can only converge if its marginals do converge we obtain

$$F_j^n(a_{n,j}x + b_{n,j}) \xrightarrow{D} G_j(x_j). \tag{2.11}$$

In particular, the sequences $\boldsymbol{a_n}$, $\boldsymbol{b_n}$ are determined by the marginal sequences $(a_i)_n, (b_i)_n$, for $i = 1, ...d$. From theorem 2.9 follows for all x_i there are γ_i such that

$$G(\infty, ..., x_i, ...\infty) = \exp\left[-(1+\gamma_i x_i)^{-1/\gamma_i}\right]. \tag{2.12}$$

To focus on the dependence structure we standardize the marginals: For all $i \in 1, ...d$ define $V_i = -(\log G_i(X_i))^{-1}$ and $U_i(t) = G_i^{\leftarrow}(e^{-1/t})$. Then all V_i are standard Fréchet distributed:

$$\begin{aligned}
\mathbb{P}(V_i \leq z_i) &= \mathbb{P}\left[-(\log G_j(X_j))^{-1} \leq z\right] \\
&= \mathbb{P}\left[X_j \leq G_j^{\leftarrow}(e^{-\frac{1}{z}})\right] \\
&= \mathbb{P}(X_j \leq U_i(x_i)) = e^{-\frac{1}{z}},
\end{aligned} \tag{2.13}$$

where $G_i^{\leftarrow}$ is the inverse of the ith margin of G and $z > 0$. Now, we can define the distribution function of the standardized random vector $\boldsymbol{V} = (V_1, ..., V_d)$:

$$\mathbb{P}(\boldsymbol{V} \leq \boldsymbol{z}) = G_0(\boldsymbol{z}) = G(U_1(z_1), ..., U_d(z_d)) \tag{2.14}$$

To gain a better understanding of the dependence structure we first have to show that each MEVD G is in its own domain of attraction:

Proposition 2.13.

1. *G is in its own domain of attraction.*

2. *G_0 is in its own domain of attraction with* $\lim_{k\to\infty} G_0^k(k\boldsymbol{z}) = G_0(\boldsymbol{z})$

Proof. The first part of proposition 2.13 is sketched in [Bei+04] chapter 8.2.1 and the second part is stated in [Bei+04] chapter 8.2.2 but a proof is omitted. For illustrative purposes, we provide a proof for the second part.

From the first statement follows there exist sequences $\boldsymbol{a}_n = (a_{i,n})_{i=1,...d} > 0$ and $\boldsymbol{b}_n = (b_{i,n})_{i=1,...d} \in \mathbb{R}^d$ with $n \in \mathbb{N}$ such that

$$\lim_{n\to\infty} G^n(\boldsymbol{a}_n\boldsymbol{x} + \boldsymbol{b}_n) = G(\boldsymbol{x}). \tag{2.15}$$

We know that we can determine these values from the convergence of the margins. Using theorem 2.9 we get

$$U_i(z_i) = \frac{z_i^{\gamma_i} - 1}{\gamma_i}. \tag{2.16}$$

Again with theorem 2.9 we can show that for all $n \in \mathbb{N}_+$ we can choose

$a_{i,n} = n^{\gamma_i}$ and $b_{i,n} = U_i(n)$ such that

$$\begin{aligned}
\log G_i^n(a_{i,n}x + b_{i,n}) &= \log G_i^n(n^{\gamma_i}x_i + U_i(n)) \\
&= -n\left[1 + \gamma_i\left(n^{\gamma_i}x + \frac{n^{\gamma_i} - 1}{\gamma_i}\right)\right]^{-1/\gamma_i} \\
&= -(1 + \gamma_i x)^{-1/\gamma_i}. \\
&= \log G_i(x_i).
\end{aligned} \tag{2.17}$$

In particular,

$$\lim_{n\to\infty} G_i^n(n^{\gamma_i}x + U_i(n)) = G(x). \tag{2.18}$$

Inserting these values for $a_{i,n}$ and $b_{i,n}$ in the following equation we get

$$\frac{U_i(nz_i) - b_{i,n}}{a_{i,n}} = U_i(z_i). \tag{2.19}$$

For $i = 1, ...d$ let $x_i = \frac{U_i(nz_i) - b_{i,n}}{a_{i,n}}$. Then plug these x_i into Eq. (2.10) and we get

$$\begin{aligned}
\lim_{n\to\infty} G_0^n(\boldsymbol{z}) &= \lim_{n\to\infty} G^n(U_1(nz_1), ..., U_d(nz_d)) \\
&= \lim_{n\to\infty} G^n(\boldsymbol{a_n x} + \boldsymbol{b_n}) = G(\boldsymbol{x}) \\
&= G_0(\boldsymbol{z}).
\end{aligned} \tag{2.20}$$

□

This characterizing property of extreme value distributions can be used to introduce the exponent measure through which we can define the spectral measure that gives us some intuition about the dependence behavior of a multivariate extreme value distribution.

Because of too many possible dependence structures an MEVD cannot be represented as a parametric family indexed by a finite-dimensional parameter vector. Nonetheless, we can pose some characterizing constraints on the possible dependence structures. As a consequence of proposition 2.13 [BR77] showed that every MEVD is described by its exponent measure:

Theorem 2.14. *There exist finite measure $\mu_1, ..., \mu_n$, and μ defined on all Borel sets A of $\mathbb{R}_+^d$ with $\inf_{(x_1,...,x_d)\in A} \max(x_1, ..., x_d) > 0$ such that*

$$\begin{aligned}
\mu_n\left\{(x_1, ..., x_d) \in \mathbb{R}_+^d : \bigvee_{i=1}^{d} x_i > z_i\right\} &= n(1 - G_0(nz_1, ..., nz_d)) \\
\mu\left\{(x_1, ..., x_d) \in \mathbb{R}_+^d : \bigvee_{i=1}^{d} x_i > z_i\right\} &= -\log G_0(z_1, ..., z_d)
\end{aligned} \tag{2.21}$$

and for each Borel set $A \subseteq \mathbb{R}_+^d$ *with* $\inf_{(x_1,...,x_d)\in A} \max(x_1, ..., x_d) > 0$ *and* $\mu(\partial A) = 0$,

$$\lim_{n\to\infty} \mu_n(A) = \mu(A). \tag{2.22}$$

Proof. In [HF06] theorem 6.1.5 the authors prove theorem 2.14 for a distribution function F in the domain of attraction of G. However, since G is in its own domain of attraction (see proposition 2.13) the proof works analog. □

The exponent measure satisfies the following homogeneity property (see [HF06] theorem 6.1.9):

Proposition 2.15. *For all* $t \in \mathbb{R}_+$ *and Borel sets in which* μ *is defined on:*

$$\mu(tA) = t^{-1}\mu(A) \tag{2.23}$$

Analog to [Bei+04] we can use the homogeneity property to decompose the exponent measure into a product measure of a simple radial and an angular part. Let $S^{d-1} = \{\boldsymbol{\omega} \in \mathbb{R}^d : ||\boldsymbol{\omega}||_1 = 1\}$ be the unit simplex with respect to the sum-norm $||(\omega_1, ..., \omega_d)||_1 = |\omega_1| + ... + |\omega_d|$. Define the bijective mapping

$$\begin{aligned} &T : \mathbb{R}^d \setminus \{\mathbf{0}\} \to (0, \infty) \times S^{d-1} \\ &T : \boldsymbol{z} \to (||\boldsymbol{z}||_1, \boldsymbol{z}/||\boldsymbol{z}||_1) \end{aligned} \tag{2.24}$$

Next, we define the spectral measure Φ on the Borel sets $B \subseteq S^{d-1} \cap [\mathbf{0}, \boldsymbol{\infty}) =: S_+^{d-1}$

$$\Phi(B) = \mu\{\boldsymbol{z} \in [\mathbf{0}, \boldsymbol{\infty}) : ||\boldsymbol{z}||_1 \geq 1, \boldsymbol{z}/||\boldsymbol{z}||_1 \in B\}. \tag{2.25}$$

The homogeneity property 2.15 of the measure μ implies

$$\mu\{\boldsymbol{z} \in [\mathbf{0}, \boldsymbol{\infty}) : ||\boldsymbol{z}||_1 \geq r, \boldsymbol{z}/||\boldsymbol{z}||_1 \in B\} = r^{-1}\Phi(B) \tag{2.26}$$

for $0 < r < \infty$. Hence, μ factors into two measures, one with density $r^{-2}dr$ and the spectral measure Φ, first introduced by [DR77]. Particularly, since the exponent measure completely describes the dependence structure of an MEVD and the spectral measure describes the exponent measure we can describe the dependence structure of an MEVD through the spectral measure.

Remark 2.16. The sum-norm used to decompose the spectral measure can usually be exchanged for other norms on $\mathbb{R}^d$. However, upper bounds do change for different norms and are not always determinable independently from G (see remark 2.19). Based on the exponent measure, different properties about the dependence structure follow like convexity properties and upper bounds (for more about this topic see [Bei+04] and [HF06]).

2.4 The Extreme Value Index

Finally, we propose the Extreme Value Index to Compare Image Distributions (EVI) based on the support of Φ to distinguish between the two distributions $\mathbb{P}_G$ and $\mathbb{P}_R$. (see problem stated in 2.2). We can canonically partition S_+^{d-1} into its so called "faces": For $\alpha \subseteq \{1,2,...,d\}$, $\alpha \neq \emptyset$, define $\Omega_\alpha := \{\boldsymbol{\omega} \in S_+^{d-1} : \forall i \in \alpha\ \omega_i > 0, \forall i \notin \alpha\ \omega_i = 0\}$. In particular, these sets are disjoint. Therefore, we can write

$$\Phi(\cdot) = \sum_{\alpha \subseteq \{1,...,d\}} \Phi_\alpha(\cdot) \tag{2.27}$$

where $\Phi_\alpha(\cdot)$ is the exponent measure restricted on Ω_α. Now, let Φ and Ψ be the exponent measures corresponding to the distributions of the random vectors $\boldsymbol{S_G}$, and $\boldsymbol{S_R}$ of the real and generated images, respectively. Define $\Phi_\alpha = \Phi(\Omega_\alpha)$ and $\Psi_\alpha = \Psi(\Omega_\alpha)$. We propose the following indicator to distinguish between $\mathbb{P}_R$ and $\mathbb{P}_G$:

Definition and Proposition 2.17. *Let $M(\Phi) := \{\alpha \subseteq \{1,...,d\} : \Phi_\alpha > 0\}$ and define the coefficients:*

$$\begin{aligned} I &= d^{-1} \sum_{\alpha \subseteq M(\Phi)\cap M(\Psi)} \frac{\Psi_\alpha}{\Phi_\alpha}(2\Phi_\alpha - \Psi_\alpha) \\ G &= d^{-1} \sum_{\alpha \subseteq M(\Psi)\setminus M(\Phi)} \Psi_\alpha \\ R &= d^{-1} \sum_{\alpha \subseteq M(\Phi)\setminus M(\Psi)} \Phi_\alpha \end{aligned} \tag{2.28}$$

Consider the function $I(\Psi) := \sum_{\alpha \subseteq M(\Phi)\cap M(\Psi)} \frac{\Psi_\alpha}{\Phi_\alpha}(2\Phi_\alpha - \Psi_\alpha)$. Then

1. *The function $I(\Psi)$ takes on its global maximum if and only if $\Phi_\alpha = \Psi_\alpha$ for all $\alpha \subseteq M(\Phi) \cap M(\Psi)$.*
2. *$I \in (-\infty, 1]$ with $I = 1$ if and only if I takes on its global maximum and $M(\Phi) = M(\Psi)$.*
3. *If there are spectral measures Ψ and $\tilde{\Psi}$ with $M(\Psi) \subseteq M(\tilde{\Psi})$ and for all $\alpha \subseteq M(\Psi) \cap M(\Phi)$ $\Psi_\alpha \leq \tilde{\Psi}_\alpha$ then $I(\Psi) \leq I(\tilde{\Psi})$.*
4. *R and G are in $[0,1]$.*

To prove (2) and (4) we first have to show the following Lemma:

Lemma 2.18. *$\Phi(S_+^{d-1}) = d$.*

Proof. [Bei+04] state this lemma on page 260 but no proof is provided. Since for $\boldsymbol{\omega} = (\omega_1, ..., \omega_d) \in S_+^{d-1}$ $||\boldsymbol{\omega}||_1 = \omega_1 + ... + \omega_d = 1$ it follows

$$\Phi(S_+^{d-1}) = \int_{S_+^{d-1}} 1 d\Phi(\boldsymbol{\omega}) = \int_{S_+^{d-1}} \omega_1 d\Phi(\boldsymbol{\omega}) + ... + \int_{S_+^{d-1}} \omega_d d\Phi(\boldsymbol{\omega}). \quad (2.29)$$

Therefore, we have to determine $\int_{S_+^{d-1}} \omega_i d\Phi(\boldsymbol{\omega})$ for $i = 1, ...d$:

$$\begin{aligned}
\int_{S_+^{d-1}} \omega_i d\Phi(\boldsymbol{\omega}) &= \int_{S_+^{d-1}} \int_{1/\omega_i}^{\infty} r^{-2} dr d\Phi(\boldsymbol{\omega}) \\
&= \int_{S_+^{d-1}} \int_{\mathbb{R}_+} \mathbb{1}_{\{r\omega_i \geq 1\}}(r\boldsymbol{\omega}) r^{-2} dr d\Phi(\boldsymbol{\omega}) \\
&= \int_{\mathbb{R}_+} \int_{S_+^{d-1}} \mathbb{1}_{\{r\omega_i \geq 1\}}(r\boldsymbol{\omega}) d\Phi(\boldsymbol{\omega}) r^{-2} dr \\
&= \int_{\mathbb{R}_+^d} \mathbb{1}_{\{z_i \geq 1\}}(\boldsymbol{z}) d\mu(\boldsymbol{z}) \\
&= \mu(\{\boldsymbol{z} \in \mathbb{R}_+^d : z_1 \geq 1\}) \\
&= -\log G_0(\infty, ..., 1, ..., \infty) \\
&= -\log G_i(U_i(1)) \\
&= 1
\end{aligned} \quad (2.30)$$

where the third equality follows from the Fubini-Tonelli theorem as Φ is defined through the finite measure μ (see theorem 2.14) and the fourth equality from definition (2.25) of Φ. From (2.29) and (2.30) follows $\Phi(S_+^{d-1}) = d$. $\square$

Now, we can turn to prove the 4 statements in proposition 2.17.

Proof. The first and the third statement follow directly from considering the function $i(x) = \frac{x}{\Phi_\alpha}(2\Phi_\alpha - x)$. $i(x)$ takes on its unique maximum Φ_α when $x = \Phi_\alpha$. Now, when we consider the function $I(\Psi)$ we see that it takes on its global maximum $d^{-1} \sum_{\alpha \subseteq M(\Phi) \cap M(\Psi)} \Phi_\alpha$ if and only if for all $\alpha \subseteq M(\Phi)$, $\Phi_\alpha = \Psi_\alpha$. The third part follows from the fact that $i(x)$ is strictly decreasing for $x \gtrless \Phi_\alpha$.

Since Φ is bounded from above so is Φ_α for $\alpha \subseteq M(\Phi)$. Therefore, it follows from the first part of the prove that $I \leq 1$ with $I = 1$ if and only if $M(\Phi) \cap M(\Psi) = M(\Phi)$ and for all $\alpha \subseteq M(\Phi)$ $\Phi_\alpha = \Psi_\alpha$ since

$$d^{-1} \sum_{\alpha \subseteq M(\Phi)} \Phi_\alpha = d^{-1} \Phi(S_+^{d-1}) = 1, \quad (2.31)$$

where the last two equalities follow from Eq. (2.27) and lemma 2.18, respectively. I is not bounded from below. Suppose $\Psi_\alpha = 1$ for some $\alpha \subseteq$

$M(\Phi) \cap M(\Psi)$, and $\Phi_\alpha \to 0$ then $i(\Psi) \to -\infty$, and as all other summands of I are 0, $I \to -\infty$.

The bounds from R, and G follow directly from lemma 2.18. $\square$

Remark 2.19. In lemma 2.18 we showed that the upper bound for Φ defined through the sum-norm is independent of G up to its dimension. However, if we define Φ through the max-norm it follows directly from the definition of Φ (2.25) and the connection from μ to G (theorem 2.14) that $\Phi(S_+^{d-1}) = -\log G_0(1, 1, ..., 1)$, hence its upper bound is not independent of G. Thus, we cannot simply exchange the sum-norm for an arbitrary norm on $\mathbb{R}^d$.

From the proposition in 2.17 we see that the coefficients I, G, and R satisfy favorable properties for interpretation. I near 1 means that the perceptual features of generated and real images can be observed with the same probability in certain subspaces of $\mathbb{R}^d$. Thus, we might interpret I near 1 as that generated and real images are qualitatively comparable and occur in the real and generated dataset with the same probability. The indicator G corresponds to artifacts produced by the generator. Hence, generated images that cannot be attributed to the real dataset. And R corresponds to modes of the real dataset that the generator does not cover, i.e. images from the real dataset that the generator cannot create. The third characteristic in 2.17 makes the EVI favorable to compare the performance of different generators. In the next section, we discuss how to empirically estimate the support of Φ.

Remark 2.20. The coefficients have to be interpreted carefully. From $I = 1$ we cannot conclude that $\mathbb{P}_G = \mathbb{P}_R$. Therefore, the introduced statistic is only a similarity indicator of $\mathbb{P}_G$ and $\mathbb{P}_R$ but it does not answer the question of whether $\mathbb{P}_G$ and $\mathbb{P}_R$ are indeed equal.

3 Support Estimation of the Exponent Measure

In this section, we discuss an empirical estimator of the EVI and its implementation. We remember the canonical partition of the unit simplex into its faces. For $\alpha \subseteq \{1, ..., d\}$, let $\alpha \neq \emptyset$, $\Omega_\alpha := \{\boldsymbol{\omega} \in S_+^{d-1} : \forall i \in \alpha \ \omega_i > 0, \forall i \notin \alpha \ \omega_i = 0\}$. Incidentally, this partition corresponds to a partition of $C := \{\boldsymbol{z} \in \mathbb{R}^d : ||\boldsymbol{z}||_1 \geq 1\}$ in $C_\alpha := \{\boldsymbol{z} \in C : \forall i \in \alpha \ z_i > 0, \forall i \notin \alpha \ z_i = 0\}$, first introduced by [GSC16]. Directly from the definition of Φ follows $\mu(C_\alpha) = \Phi(\Omega_\alpha)$. This connection leads to the multivariate support estimation algorithms DAMEX ([GSC16]) and CLEF ([CS16]).

For the following, we consider the random vector $\boldsymbol{V} = (V_1, ..., V_d)$ where each V_i $i = 1, ...d$ is standard Fréchet distributed and $\boldsymbol{V}$ has distribution G_0 belonging to a MEVD. Further, let μ be the exponent measure of G_0. The aim of this section is to estimate the probabilities $\mu(C_\alpha) = \Phi(\Omega_\alpha)$.

3.1 The DAMEX Algorithm

[GSC16] developed the Detecting anomalies among multivariate extremes (DAMEX) algorithm to cluster extreme values in a sample of random vectors. In contrast to our setting they use the max-norm to define the spectral measure. Hence, they consider the cube $\{\boldsymbol{w} \in \mathbb{R}^d : ||\boldsymbol{w}||_{\max} = 1\}$ instead of the unit-simplex. Therefore, we exchange the max-norm with the sum-norm in the DAMEX-algorihtm. Further, they assume that the distribution of the random vectors is not MEVD itself but lies in a MEVD domain of attraction. However, since every MEVD lies in its own domain of attraction (see proposition 2.13) the reasoning follows the same lines. To cluster the extreme observations they consider the set C_α. Since there are $2^d - 1$ of these sets they assume that the support of μ is sparse, in the sense of $|M| \ll 2^d - 1$ which they argue is a reasonable assumption in high dimensional settings. A consequence of theorem 2.14 is that $\lim_{k \to \infty} k\mathbb{P}(kC_\alpha) = \lim_{k \to \infty} \mu_k(C_\alpha) = \mu(C_\alpha)$. The left-hand side can then be approximated by the empirical distribution

$$\hat{\mu}_k(C_\alpha) := k\hat{P}_n(kC_\alpha) := \frac{k}{n} \sum_{i=1}^{n} \mathbb{1}\{\hat{V}_i \in kC_\alpha\}. \tag{3.1}$$

Problematically, C_α for $\alpha \neq \{1, ...d\}$ is not a continuity set of μ since it has empty interior and thus, $\lim_{k \to \infty} \mu_k(C_\alpha) = \mu(C_\alpha)$ does not necessarily hold. Therefore, [GSC16] introduce the rectangles $R_\alpha^\epsilon := \{\boldsymbol{z} \in C : \forall i \in \alpha \ z_i > \epsilon, \forall i \notin \alpha \ z_i \leq \epsilon\}$. These R_α^ϵ also partition C but in contrast to the C_α are continuity sets of μ such that $\lim_{k \to \infty} \mu_k(R_\alpha^\epsilon) = \mu(R_\alpha^\epsilon)$ holds under the assumption (2.10). Further, [GSC16] show that $\lim_{\epsilon \to 0} \mu(R_\alpha^\epsilon) = \mu(C_\alpha)$ hence for large k we might approximate

$$\mu(C_\alpha) \simeq \mu(R_\alpha^\epsilon) \approx k\hat{P}_n(kC_\alpha). \tag{3.2}$$

Concluding, the different $\mu(C_\alpha)$ are approximated and in the next step the $\mu(C_\alpha)$ not passing a threshold are set to 0. The clusters that pass the threshold are gathered into a set that describes the support of μ where the probability of values to observe is non-negligible.

This approach might fail to recover the support of μ if the observations are spread onto too many C_α sets, or rather if there are not enough observations with respect to the number of dimensions. Derived from the DAMEX-algorithm the Clustering Extreme Features (CLEF)-algorithm ([CS16]) offers another approach. CLEF is also based on EVT and aims to estimate the support of μ but drops the condition that observations have to be small simultaneously to belong to the same cluster.

3.2 CLEF Algorithm

With respect to the intuitive line of reasoning at the beginning of section 2, one might argue that observations belonging to the same class necessarily score high on the same features intrinsic to their class but might not need to score low on all other features. Since there are differences within each class one observation of class A might exhibit a feature and another observation of A not. Therefore, observation one scores high on this feature while observation two does not. However, both belong to class A. Hence, some features are not necessary for the classification of certain observations and we could assume that we observe random variables $V_i + \epsilon$ where ϵ is a random noise term that explains why observations belonging to the same class lie on different C_α. The DAMEX-algorithm would have put those two observations into two different classes. This problem CLEF tries to solve. Instead of trying to cluster the observations onto the sets C_α [CS16] propose to infer the support of μ from the extremal joint excess coefficient ρ_α. Let $\Gamma_\alpha = \{\boldsymbol{z} \in C : \forall i \in \alpha \; z_i > 1\}$ and define

$$\rho_\alpha = \mu(\Gamma_\alpha). \tag{3.3}$$

The CLEF-algorithm aims to recover the maximal elements of the set $\tilde{M} := \{\alpha \in \{1, ..., d\} : \rho_\alpha > 0\}$, hence all $\alpha \in \tilde{M}$ such that there does not exist $\beta \in \tilde{M}$ with $\alpha \subseteq \beta$. Unlike the C_α the Γ_α do not partition C. In particular, for $\alpha \subseteq \beta$ $\Gamma_\beta \subseteq \Gamma_\alpha$, thus $\mu(\Gamma_\beta) \leq \mu(\Gamma_\alpha)$. Therefore, in contrast to the DAMEX algorithm $\mu(\Gamma_\alpha)$ does not have to cross a fixed threshold, but [CS16] set $\mu(\Gamma_\alpha) > 0$ if the empirical counterpart of

$$\frac{\mu(\Gamma_\alpha)}{\mu(\Delta_\alpha)} > \kappa \tag{3.4}$$

with $\Delta_\alpha = \{\boldsymbol{z} \in C : \sum_{j \in \alpha} \mathbb{1}_{x_j \geq 1} \geq |\alpha| - 1\}$ and fixed threshold κ. With other words, there has to be a sufficient increase in mass when removing the constraint that for any $i \in \beta \subset \alpha$ $x_i > 1$.

Concerning the estimation of the EVI most importantly, [CS16] show that the maximal elements of $\tilde{M}$ and the for α-set inclusion maximal elements in M coincide.

Lemma 3.1. *For $\alpha \in \{1, ..., d\}$ α is maximal in $\tilde{M}$ $\iff$ α is maximal in M.*

From this lemma we can conclude that we can recover parts of the support of the exponent measure with the CLEF algorithm. Particularly, the elements in $\tilde{M}$ correspond to simplex-faces of S_+^{d-1} where Φ has nonzero mass.

3.3 EVI Implementation Details

With the DAMEX algorithm the implementation of the EVI is straight forward (see Algorithm 3.2). Let μ, and ν be the exponent measure of $\mathbb{P}_R$, and $\mathbb{P}_G$, respectively. The DAMEX algorithm returns the set M such that for all $\alpha \subseteq M$ $\mu_\alpha > 0$ and for each α the number of observations $|\hat{C}_\alpha|$ that lie in C_α. To ensure that the empirical estimators of the EVI score satisfy the bounds shown in proposition 2.17 we compute

$$\hat{\mu}_\alpha = d \frac{|\hat{C}_\alpha|}{\sum_{\beta \subseteq M} |\hat{C}_\beta|}. \tag{3.5}$$

Finally, we plug in this empirical estimator into the equations from definition 2.17

$$\begin{aligned}
\hat{I} &= d^{-1} \sum_{\alpha \subseteq M(\mu) \cap M(\nu)} \frac{\hat{\nu}_\alpha}{\hat{\mu}_\alpha}(2\hat{\mu}_\alpha - \hat{\nu}_\alpha) \\
\hat{G} &= d^{-1} \sum_{\alpha \subseteq M(\nu) \setminus M(\mu)} \hat{\nu}_\alpha \\
\hat{R} &= d^{-1} \sum_{\alpha \subseteq M(\mu) \setminus M(\nu)} \hat{\mu}_\alpha.
\end{aligned} \tag{3.6}$$

Algorithm 3.2.

Input: $\epsilon > 0$, threshold k, $\mu_{min} \geq 0$, generated dataset $gd = (g_{i,j})_{i=1,...n,j=1,...d}$, real dataset $rd = (r_{i,j})_{i=1,...n,j=1,...d}$

(a) standardize via marginal rank-transformation.

for i in $1...n$:

$V_i^r = (-1/\log \hat{F}_j(r_{i,j}))_{j=1,...d}$

$V_i^g = (-1/\log \hat{G}_j(g_{i,j}))_{j=1,...d}$

(b) recover M_r, M_g and each $|\hat{C}_\alpha|$, $|\hat{C}_\beta|$ with the DAMEX algorithm

M_r, $(|\hat{C}_\alpha|)_{\alpha\subseteq M_r}$ = DAMEX($\boldsymbol{V}^r$, ϵ, k, μ_{min})

M_g, $(|\hat{C}_\beta|)_{\beta\subseteq M_g}$ = DAMEX($\boldsymbol{V}^g$, ϵ, k, μ_{min})

(c) compute empirical estimator $\hat{\mu}_\alpha$, $\hat{\nu}_\beta$

for α, β in M_r, M_g:

$\hat{\mu}_\alpha = d\frac{|\hat{C}_\alpha|}{\sum_{\gamma\subseteq M_r}|\hat{C}_\gamma|}$, $\hat{\nu}_\beta = d\frac{|\hat{C}_\beta|}{\sum_{\gamma\subseteq M_g}|\hat{C}_\gamma|}$

(d) determine the sets $\mathcal{I} = M_r \cap M_g$, $\mathcal{R} = M_r \setminus M_g$, $\mathcal{G} = M_g \setminus M_r$

(e) compute $\hat{I}$, $\hat{G}$, $\hat{R}$

$\hat{I} = d^{-1}\sum_{\alpha\subseteq\mathcal{I}}\frac{\hat{\nu}_\alpha}{\hat{\mu}_\alpha}(2\hat{\mu}_\alpha - \hat{\nu}_\alpha)$

$\hat{G} = d^{-1}\sum_{\alpha\subseteq\mathcal{G}}\hat{\nu}_\alpha$

$\hat{R} = d^{-1}\sum_{\alpha\subseteq\mathcal{R}}\hat{\mu}_\alpha$

Output: $\hat{I}$, $\hat{G}$, $\hat{R}$

To infer the EVI on the basis of the CLEF algorithm we assume that our observations are confounded by noise. Hence, we observe $\boldsymbol{V_i} + \boldsymbol{\epsilon_i}$ for $i = 1, ...n$. With the CLEF algorithm we recover the set $\tilde{M}$ consisting of the maximal elements of the set M. From lemma 3.1 it follows that for all $\alpha \subseteq \tilde{M}$ $\mu_\alpha > 0$. Therefore, we assume that if an observation $\boldsymbol{V_i} + \boldsymbol{\epsilon_i} \in \Gamma_\alpha$ the observation $\boldsymbol{V_i}$ actually lies in C_α. Problematically, the Γ_α are not disjoint and therefore, we have to remove observations we have already used to estimate another support (see Algorithm 3.3).

Algorithm 3.3.

Input: threshold k, $kappa > 0$, generated dataset $gd = (g_{i,j})_{i=1,\dots n, j=1,\dots d}$, real dataset $rd = (r_{i,j})_{i=1,\dots n, j=1,\dots d}$

(a) standardize via marginal rank-transformation

for i in $1 \dots n$:

$V_i^r = (-1/\log \hat{F}_j(r_{i,j}))_{j=1,\dots d}$

$V_i^g = (-1/\log \hat{G}_j(g_{i,j}))_{j=1,\dots d}$

(b) recover $\tilde{M}_r$, $\tilde{M}_g$ with the CLEF algorithm

$\tilde{M}_r = \mathrm{CLEF}(V^r, k, \kappa)$, $\tilde{M}_g = \mathrm{CLEF}(V^g, k, \kappa)$

(c) gather the sets $|\hat{C}_\alpha|$, $|\hat{C}_\beta|$

for α, β in $\tilde{M}_r$, $\tilde{M}_g$:

count $V_i^r \in k\Gamma_\alpha$, compute $\hat{C}_\alpha$, remove counted V_i^r

count $V_i^g \in k\Gamma_\alpha$, compute $\hat{C}_\beta$, remove counted V_i^g

(d) compute empirical estimator $\hat{\mu}_\alpha$, $\hat{\nu}_\beta$

for α, β in M_r, M_g:

$\hat{\mu}_\alpha = d\frac{|\hat{C}_\alpha|}{\sum_{\gamma \subseteq M_r} |\hat{C}_\gamma|}$, $\hat{\nu}_\beta = d\frac{|\hat{C}_\beta|}{\sum_{\gamma \subseteq M_g} |\hat{C}_\gamma|}$

(e) determine the sets $\mathcal{I} = M_r \cap M_g$, $\mathcal{R} = M_r \setminus M_g$, $\mathcal{G} = M_g \setminus M_r$

(f) compute $\hat{I}$, $\hat{G}$, $\hat{R}$

$\hat{I} = d^{-1} \sum_{\alpha \subseteq \mathcal{I}} \frac{\hat{\nu}_\alpha}{\hat{\mu}_\alpha}(2\hat{\mu}_\alpha - \hat{\nu}_\alpha)$

$\hat{G} = d^{-1} \sum_{\alpha \subseteq \mathcal{G}} \hat{\nu}_\alpha$

$\hat{R} = d^{-1} \sum_{\alpha \subseteq \mathcal{R}} \hat{\mu}_\alpha$

Output: $\hat{I}$, $\hat{G}$, $\hat{R}$

Remark 3.4. In algorithm 3.3 we assign the observations somewhat randomly to a set C_α possibly influencing inference. In section 5 we propose ideas to improve the the EVI estimation.

4 Tests

In this section, we test whether the EVI is aligned with perceived image quality and diversity and whether it can differentiate between a decline of quality or diversity. Based on the line of reasoning from section 2.2 we can use the maximum function to summarize the information of a feature-layer. This enables us to compute MPIs like the FID, Precision and Recall in different layers of a CNN. Activations of different layers, particularly of more shallow layer, might be more independent of the training data. Therefore, using different feature-layer we might escape the problems associated with the penultimate layer of the InceptionNet that is usually used to compute FID, Precision and Recall. [Zha+18] show that features of the VGG network ([SZ15]) pretrained on ImageNet correspond particularly well with human perception. For this reason, we compute the EVI on four feature-layer of the VGG-16 network. Each time we choose the activations before the max-pooling layer such that a max-pooling layer is scheduled afterwards (for the actual implementation of the feature embeddings based on the feature-layer see Appendix 6.5). As the activations of the InceptionNet are commonly used for MPIs like the FID, Precision, and Recall we also compute the EVI for the activations of four feature-layer within the IncpetionNet. For each feature-layer of the VGG network and the InceptionNet we also compute FID, Precision and Recall.

Since for many experimental settings, the DAMEX algorithm failed to recover a support, we use the CLEF algorithm to recover the support of the exponent measure. Therefore, we use algorithm 3.3 to compute the EVI for all following results.

For the following experiments, we consider the publicly available Flickr-Faces-HQ (FFHQ) dataset, a high-quality image dataset of human faces, originally created as a benchmark for generative adversarial networks ([KLA19]), a subset of Extreme UltraViolet (EUV) solar images from the SDO AIA instrument ([Lem+12]), and a subset of 50 randomly chosen ImageNet classes ([Den+09]). The script to download and create each dataset is appended (see Appendix 6.5). We use the publicly available pretrained StyleGAN2-Ada generator ([Kar+20]) to generate FFHQ images and a pretrained Projected GAN generator ([Sau+21]) to generate EUV solar images ([Che+23], for the specific generator network see Appendix 6.5).

To fix the hyperparameters k and κ used by the CLEF algorithm (see algorithm 3.3 and section 3.2), we conducted a hyperparameter search for each dataset.

4.1 Hyperparameter

We computed the number of supports and the discarded observations for thresholds ranging from 10, 11, 12, 13, 14, and 15 and kappa 0.2 or 0.3.

Tables for all combinations can be found in Appendix 6.1. We have to acknowledge a tradeoff between an increase in the threshold parameter and observations that could not be associated with a recovered support. On the one hand we want to use a large threshold to improve the approximation of the exponent measure. However, discarding too many observations would lead to a heavily biased sample. Particularly, for the hyperparameter tests of the InceptionNet we observe for all datasets that less observations are discarded and more clusters are found when we recover the supports of the activation distributions of feature-layer deeper within the neural network (see Table 1 and Table 3). This observation might mean that the InceptionNet indeed partitions its activations into different hyperplanes. Furthermore, deeper feature-layer should be better at distinguishing between different features as would correspond to the higher number of clusters. In future research, labeled datasets could be used to see whether these partitions correspond to cluster of observations with the same label. However, since classes often share features a definite clustering might not be possible. A similar observation applies for the deeper feature-layer of the VGG network but not as definite as for the IncpetionNet feature-layer (see Table 2 and Table 4).

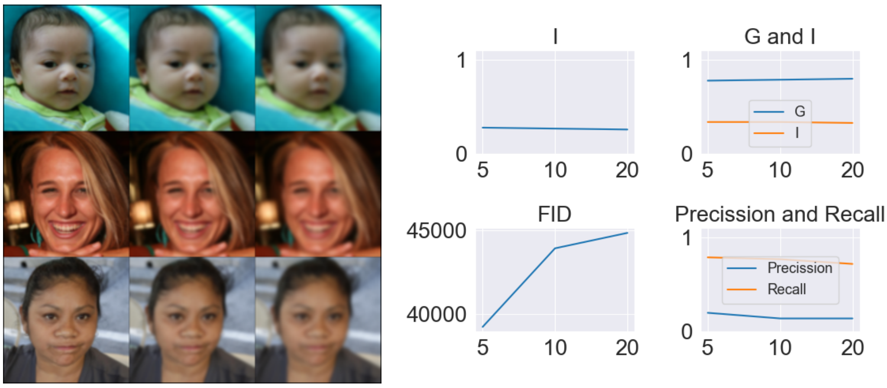
For the following experiments, we choose hyperparameter combinations such that the threshold parameter is maximal while discarding less than a quarter of the observations (for the exact hyperparamter combinations per feature-layer see Appendix 6.5). Since in the more shallow feature-layer many observations are discarded we refrained from computing the experiments for the first feature-layer of both the InceptionNet and the VGG network.

We have to note that the support estimation depends heavily on the hyperparameter choice and the sample size indicating that the observations do not exhibit a definite cluster behavior. This hyperparameter dependence obviously impacts the consistency of the EVI.

4.2 Quality Test

To measure how the EVI corresponds to a loss of image quality we fix the diversity of our test dataset. For this purpose, we compare two versions of the same dataset, one being distorted by a Gaussian filter with increasing standard deviation. We expect that the I score decreases with increasing standard deviation. The G score, which should measure artifacts, should be higher than the R score and increase with increasing standard deviation. The R score should be relatively small in comparison since the distorted dataset still should be as diverse as the original dataset.

In Tables 7 and 8 we see the results for the FFHQ dataset. As expected the FID increases with increasing standard deviation over all feature-layer. In most feature-layer the I score can capture a decline in image quality. However, the indicators sensitivity is small. The G and R score like Precision and Recall show that quality is worse in comparison to mode coverage but



(a) (b)

Figure 2: a) shows from left to right increasingly blurred FFHQ faces. b) shows the model performance indices computed for the activations of 50000 increasingly blurred FFHQ images on the InceptionNet's second feature-layer.

neither score pair is sensitive to the decrease in image quality. (An example is shown in Figure 2).

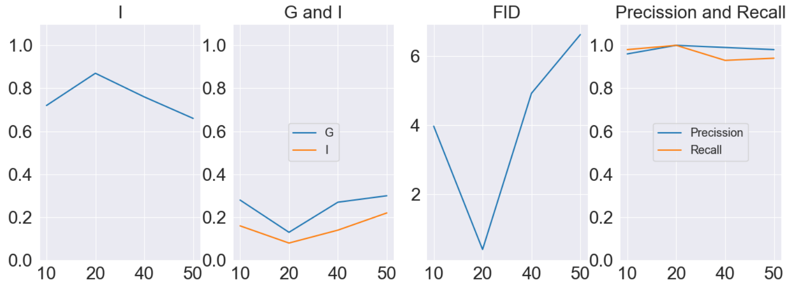
The experiments for the ultra-violet solar dataset (Tables 9 and 10) turned out similar but for the EVI score computed VGG-16 activations and Precision and Recall computed on activations of both networks. All these scores could not determine an intersecting support and therefore the G and R scores are almost 1 and the Precision and Recall score almost 0 for all feature-layer.

Neither the EVI score nor Precision and Recall could consistently and sufficiently show that the image quality declines.

4.3 Diversity Tests

Next, we fix the quality of our dataset and reduce the number of different image classes to test whether the EVI can measure a loss of diversity when we fix the dataset's quality. We compare a subset of 25 ImageNet classes against subsets of 10, 20, 40, 50 classes (see Appendix for which classes we have used). Since the CLEF algorithm is highly dependent on the sample size we use 25000 observations for each experiment. Thus, for the 25 classes subset each class consists of 400 observations while the 10 classes subset of 1000, the 20 classes subset of 500, and so forth. As we use only half the observations, we halve the threshold used recommended by the hyperparameter tests.

We expect I to peek for 20 subsets and decrease for less and more subsets than 20. More interestingly, G should start to increase when we use more than 20 subsets while R should only be decreasing from 10 to 20 subsets. However, in Tables 11 and 12 we observe that both G and R are correlated



(a)

(b)

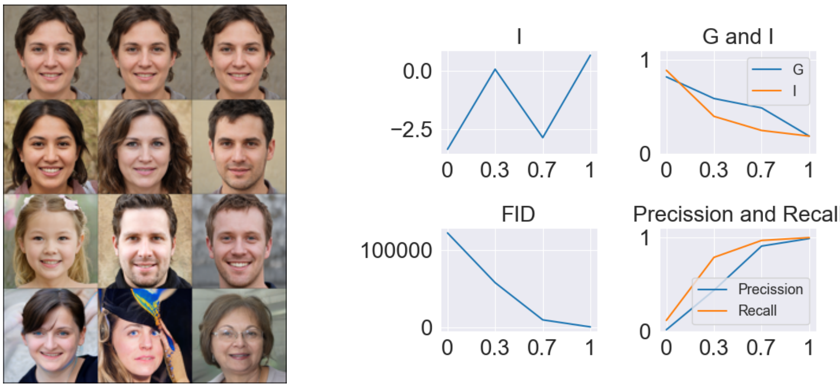
Figure 3: a) shows the model performance indices computed for different subsets of 25000 ImageNet images on the InceptionNet's second feature-layer. b) shows 20 images from 20 randomly picked ImageNet classes.

(exemplified in Figure 3). One possible explanation is that the ImageNet classes are quite diverse within each class. Since we use more observations per class for the subset with 10 classes, the EVI recognizes the additional observations as generated artifacts and when we consider a set with more classes but fewer images then these missing images are interpreted as missed diversity in the sample. However, the I score behaves as expected. In contrast, the Precision and Recall score soon saturates and differences are hard to interpret. However, the small drop in recall might be due to the same reasoning to the increase of R.

4.4 Quality for Diversity Tradeoffs

In this section, we compare the FFHQ dataset against StyleGAN2-Ada ([Kar+20]) generated images and ultra-violet solar images against Projected GAN ([Sau+21]) generated images with different levels of truncation. Truncation provides a tradeoff between perceptual quality and variation ([KLA19]). With an increase of the truncation factor the samples become more diverse but their quality declines. Hence, we would expect that G increases with increasing truncation in contrast to R which should decrease.

Based on Tables 13, 14, 15, and 16 and exemplified in Figure 4 we observe that neither G nor Precision meet their expectations. A possible explanation is that observations of a diverse dataset scatter over too many supports such that the CLEF algorithm cannot recover sufficient mass to determine any



(a) (b)

Figure 4: a) shows generated FFHQ faces with truncation parameter varying form 0, 0.3, 0.7, 1 from top to bottom. While the images at the top show very little variations the images at the bottom vary a lot but exhibit clear artifacts. b) shows correspondingly the model performance indices computed for InceptionNet's second feature-layer activations of 50000 generated FFHQ images.

supports. In contrast, CLEF find only a few supports with much mass for the strongly truncated dataset. Then, possibly only a few to zero supports intersect leading to a high G score.

In this setup [Kyn+19] have already shown that FID puts more weight on diversity than quality. Here we provide further evidence that also for a dataset with a different domain than the FFHQ dataset FID improves steadily as diversity increases even though images exhibit artifacts (see Figure 4 and Figure 5).

5 Conclusion

In this work we made a first attempt to model the activation distributions of deep convolutional networks with Extreme Value Theory. We observed that looking ever deeper within neural networks the clustering behavior of maximum values of activation distributions becomes more distinct. Further, we introduced the Extreme Value Index to Compare Distributions, a model performance index to measure diversity and quality of a generated dataset. The proposed score could not consistently measure variations in diversity and quality of image datasets. However, there are some modifications that might improve the EVI scores.

We rather arbitrarily assign an observation to a set C_α when estimating the EVI based on the CLEF algorithm 3.3. An intuitive improvement might be to assign each observation to its nearest set C_α after CLEF has determined which C_α support the spectral measure. Thus, for each observation $\boldsymbol{V}$ we might determine the distance D_α to all planes C_α and assign $\boldsymbol{V}$ to the set C_α for which D_α is minimal.

Right now, we cluster observations onto subspaces of a hyperplane although it might be preferable to cluster attributes of observations. Instead of taking the maximum over all the activations, we may take the maximum over patches of the feature-layer. Hereby, we increase the sample size to improve the approximation of the exponent measure while possibly maintaining more information.

Currently, we empirically fix hyperparameter combinations to use the CLEF-algorihtm. [GSC17] derive upper bounds for $|\hat{\mu}_k(C_\alpha) - \mu(C_\alpha)|$ based on the definition of $\hat{\mu}_k(C_\alpha)$ in Eq. (3.1). Since we need $\hat{I}$, $\hat{R}$, and $\hat{G}$ to satisfy the bounds shown in proposition 2.17 a proof for the upper bounds of $|\hat{I} - I|$, $|\hat{G} - G|$, and $|\hat{R} - R|$ needs to take this constraint into account. We might derive a suitable threshold parameter considering these upper bounds. Further, [CSS18] propose to base the parameter κ on the hypothesis that a multivariate version of the coefficient of Ledford and Tawn ([lT96]) and Ramos and Ledford ([RL09]) is one.

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6 Appendix

6.1 Hyperparameter

Tables 1, 2, 3, 4, 5, and 6 show the hyperparameter tests for the CLEF algorithm. Each table presents the hyperparameter tests for the four feature-layer of the InceptionNet-V3 and the VGG-16 network. The tables show the number of recovered clusters ("number of clusters") by the CLEF algorithm and the number of observations which could not be assigned to a cluster ("discarded observations") for each combination of chosen threshold and kappa.

6.2 Quality Tests

Tables 7, 8, 9, and 10 show the Gaussian blur tests described in section 4.3. Each table presents the I, G, R scores, FID, Precision, and Recall for three feature-layer of the InceptionNet-V3 and the VGG-16 network.

6.3 Diversity Test

Tables 11 and 12 show the ImageNet subset tests described in section 4.3. Each table presents the I, G, R scores, FID, Precision, and Recall for three feature-layer of the InceptionNet-V3 and the VGG-16 network.

6.4 Quality for Diversity Tradeoffs

Tables 13, 14, 15, and 16 show the ImageNet subset tests described in section 4.3. Each table presents the I, G, R scores, FID, Precision, and Recall for three feature-layer of the InceptionNet-V3 and the VGG-16 network.

6.5 Code

The appended USB stick contains:

- a folder "Create_Datasets" with Python scripts to create the FFHQ, solar image, and the used ImageNet subset.
- a folder "MPI_Score" with Python scripts used to compute the tests described in section 4.1, 4.2, 4.3, and 4.4. The file "MPI_Score/main.py" contains the hyperparameters used for each test and the file "MPI_Score/MPI_Score.py" a Python implementation of algorithm 3.3.
- a folder "embed_features" with a Python script to load the used feature-layer of the InceptionNet-V3 and VGG-16. Additionally, the Python script to embed the datasets into the feature-layer is provided.

- a folder "networks" containing the pretrained networks to generate FFHQ images and solar images.

6.6 Ultra-violet solar images

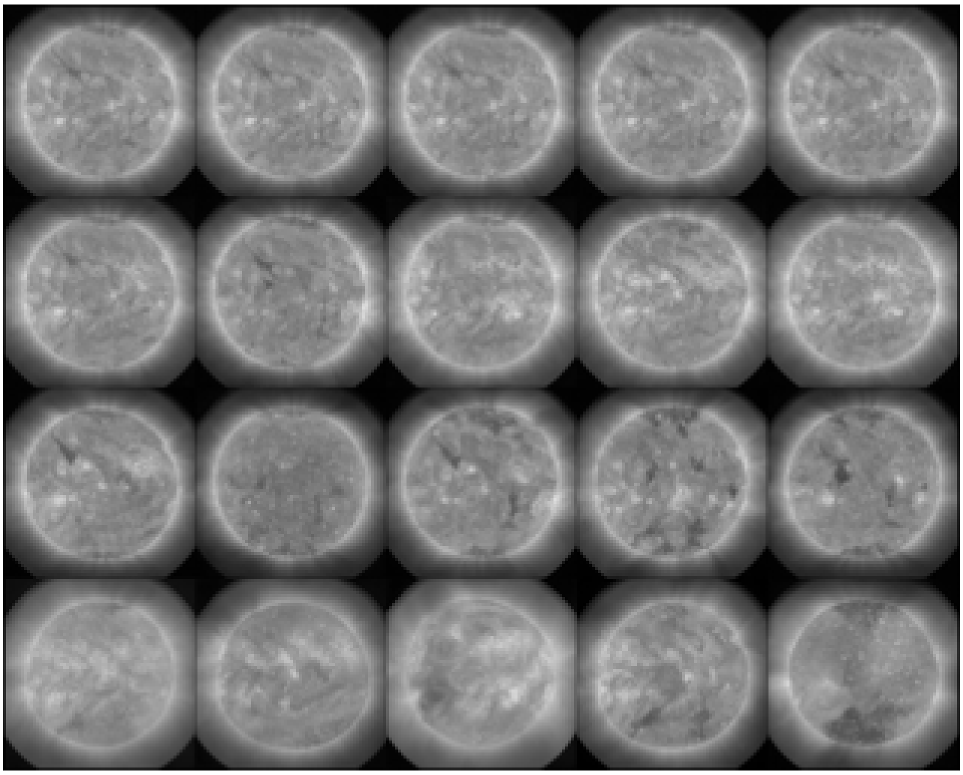

Figure 5: This Figure depicts generated ultra-violet solar images with truncation parameter varying form 0, 0.3, 0.7, 1 from top to bottom. While the images at the top show very little variations the images at the bottom vary a lot but exhibit clear artifacts (for example, they loose the round shape of the sun).

Layer1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	333	42	291	35	259	26	228	19	208	18	189	15
discarded observations	22240	30130	24380	32577	26031	35436	27620	37350	29062	38305	30397	39815

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	2424	639	2222	559	2017	489	1816	438	1676	373	1536	326
discarded observations	13355	16618	14864	18299	16325	19956	17621	21618	18806	23338	19936	24877

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	2035	760	1807	651	1672	579	1507	501	1377	438	1251	383
discarded observations	8787	13829	10116	15492	11392	17372	12681	18923	13796	20638	14954	21951

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	2588	1332	2483	1213	2404	1144	2273	1035	2126	956	2002	857
discarded observations	2185	9273	3203	11630	4499	13890	5689	15996	6878	19115	8219	21133

Table 1: Hyperparameter tests for the four feature layers of the InceptionNet-V3 given 50000 FFHQ images.

Layer 1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	1259	107	1071	89	936	72	819	57	737	53	655	48
discarded observations	15239	23938	17468	26538	19260	28646	21048	31147	22916	32463	24449	33873

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	773	24	560	19	404	16	301	12	243	8	192	7
discarded observations	12305	31005	14639	34117	16930	37115	19255	39812	21303	41651	23506	42243

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	659	49	504	40	405	33	338	31	301	24	248	22
discarded observations	8277	23465	10304	26666	12485	29926	14363	31373	15999	33973	18191	35219

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	245	17	191	15	156	14	126	13	108	10	98	7
discarded observations	12882	32604	15785	34608	18259	35733	21033	37015	22987	40491	25133	43487

Table 2: Hyperparameter tests for the four feature layers of the VGG-16 given 50000 FFHQ images.

Layer1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	74	26	46	25	33	24	31	24	30	24	27	21
discarded observations	28966	37906	32851	38776	37549	39525	39363	40211	40315	40825	41225	42281

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	812	547	816	529	811	498	797	485	788	455	777	442
discarded observations	18724	23949	20434	25465	21740	27583	23020	28746	24387	30763	25908	31521

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	995	590	951	570	906	546	870	519	834	514	814	502
discarded observations	15133	18961	16187	20747	17215	22938	18225	24862	19427	25770	20480	26726

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	1587	859	1581	804	1616	761	1601	704	1564	649	1557	626
discarded observations	11361	13533	11422	14551	11571	16202	11845	17432	12148	18746	12664	20271

Table 3: Hyperparameter tests for the four feature layers of the InceptionNet-V3 given 38676 extreme ultra-violet solar images.

Layer1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	130	57	123	61	117	63	119	62	115	63	121	62
discarded observations	29339	35660	31581	36683	33674	37506	34676	38268	35642	38979	36326	39565

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	304	119	293	116	278	121	271	118	255	114	247	113
discarded observations	22185	31560	24743	33377	27603	34119	29774	35172	30972	35934	32074	36623

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	840	398	794	390	761	378	733	361	720	360	696	349
discarded observations	14355	22393	16098	24272	17675	25442	18927	26566	20116	27721	21230	28701

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	1402	1214	1329	1113	1316	1115	1271	1092	1214	1020	1185	895
discarded observations	13902	20845	15094	22210	16291	24196	17645	26097	18932	28259	20197	30017

Table 4: Hyperparameter tests for the four feature layers of the VGG-16 network given 38676 extreme ultra-violet solar images.

Layer1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	141	29	119	27	104	17	89	13	81	11	72	10
discarded observations	22718	32988	25107	34348	27269	37236	29429	40721	30722	42230	32239	42798

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	153	11	112	8	87	7	73	5	58	5	50	4
discarded observations	14993	33194	17785	37112	21096	39529	23215	42152	25737	42670	27440	44343

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	34	4	26	2	25	2	22	1	19	1	19	1
discarded observations	23189	43784	28769	47074	30660	47369	33300	48059	35722	48229	36662	48367

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	1273	457	1186	440	1121	412	1051	390	998	365	933	341
discarded observations	3341	24551	5102	27132	6864	29296	8332	31104	9976	31224	11722	33030

Table 5: Hyperparameter tests for the four feature-layer of the InceptionNet-V3 given 50000 ImageNet images.

Layer1

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	141	29	119	27	104	17	89	13	81	11	72	10
discarded observations	22718	32988	25107	34348	27269	37236	29429	40721	30722	42230	32239	42798

Layer 2

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	153	11	112	8	87	7	73	5	58	5	50	4
discarded observations	14993	33194	17785	37112	21096	39529	23215	42152	25737	42670	27440	44343

Layer 3

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	34	4	26	2	25	2	22	1	19	1	19	1
discarded observations	23189	43784	28769	47074	30660	47369	33300	48059	35722	48229	36662	48367

Layer 4

threshold	10		11		12		13		14		15	
kappa	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.3
number of cluster	1273	457	1186	440	1121	412	1051	390	998	365	933	341
discarded observations	3341	24551	5102	27132	6864	29296	8332	31104	9976	31224	11722	33030

Table 6: Hyperparameter tests for the four feature-layer of the VGG-16 network given 50000 ImageNet images.

	Layer 2			Layer 3			Layer 4		
Blur	5	10	20	5	10	20	5	10	20
I	-1.02	-1.74	-1.90	0.28	0.27	0.26	0.31	0.26	0.30
G	0.46	0.49	0.49	0.78	0.79	0.80	0.63	0.64	0.64
R	0.46	0.48	0.47	0.34	0.34	0.33	0.39	0.40	0.39
FID	83xe4	93xe4	95xe4	39xe3	43xe3	44xe3	3811.1	4177.6	4240.7
Precission	0.06	0.04	0.04	0.20	0.14	0.14	0.42	0.30	0.32
Recall	0.32	0.26	0.24	0.79	0.77	0.72	0.81	0.80	0.75

Table 7: Gaussian blur tests for three feature-layer of the InceptionNet-V3 given 50000 FFHQ images.

	Layer 2			Layer 3			Layer 4		
Blur	5	10	20	5	10	20	5	10	20
I	0.02	0.02	0.02	0.17	0.14	0.11	0.14	0.11	0.11
G	0.84	0.85	0.86	0.86	0.88	0.86	0.87	0.88	0.89
R	0.75	0.75	0.76	0.66	0.70	0.71	0.63	0.64	0.68
FID	26xe6	29xe6	29xe6	54xe5	60xe5	61xe5	25xe4	28xe4	28xe4
Precission	0.01	0.00	0.00	0.04	0.02	0.02	0.09	0.05	0.05
Recall	0.10	0.08	0.07	0.20	0.15	0.16	0.16	0.12	0.11

Table 8: Gaussian blur tests for three feature-layer of the VGG-16 network given 50000 FFHQ images.

	Layer 2			Layer 3			Layer 4		
Blur	5	10	20	5	10	20	5	10	20
I	-0.01	-0.09	-0.20	0.26	0.16	0.12	-0.13	-1.22	-0.77
G	0.46	0.45	0.45	0.61	0.69	0.69	0.86	0.86	0.86
R	0.44	0.49	0.50	0.47	0.51	0.52	0.74	0.73	0.74
FID	53053	60594	62011	4353.0	4743.0	4821.0	1451.0	1508.0	1520.0
Precission	0.00	0.00	0.00	0.02	0.02	0.02	0.01	0.01	0.01
Recall	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01

Table 9: Gaussian blur tests for three feature-layer of the InceptionNet-V3 given 38676 ultra-violet solar images.

	Layer 2			Layer 3			Layer 4		
Blur	5	10	20	5	10	20	5	10	20
I	-0.02	-0.02	0.00	-0.56	-0.22	-0.21	-0.14	-0.05	-0.03
G	0.96	0.99	0.99	0.95	0.97	0.97	0.95	0.97	0.97
R	0.83	0.87	0.87	0.93	0.93	0.93	0.95	0.93	0.93
FID	50xe5	56xe5	57xe5	87xe4	10xe5	10xe5	19701	21442	21637
Precission	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Recall	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 10: Gaussian blur tests for three feature-layer of the VGG-16 network given 38676 ultra-violet solar images.

	Layer 2				Layer 3				Layer 4			
Subset	10	20	40	50	10	20	40	50	10	20	40	50
I	0.60	0.84	0.58	0.62	0.72	0.87	0.76	0.66	0.32	0.54	0.32	0.29
G	0.27	0.14	0.38	0.37	0.28	0.13	0.27	0.30	0.73	0.40	0.59	0.65
R	0.25	0.12	0.18	0.17	0.16	0.08	0.14	0.22	0.53	0.25	0.48	0.53
FID	1.91	0.18	4.57	6.12	3.97	0.40	4.92	6.62	105.18	9.88	44.3	59.2
Precission	0.97	1.00	0.99	0.99	0.96	1.00	0.99	0.98	0.55	0.94	0.99	0.98
Recall	0.99	1.00	0.93	0.93	0.98	1.00	0.93	0.94	0.97	1.00	0.76	0.75

Table 11: ImageNet subset tests for three feature-layer of the InceptionNet-V3 given 25000 ImageNet images.

	Layer 2				Layer 3				Layer 4			
Subset	10	20	40	50	10	20	40	50	10	20	40	50
I	0.45	0.69	0.46	0.48	-0.09	0.60	-0.05	-0.70	0.23	0.63	-0.01	0.13
G	0.43	0.14	0.64	0.64	0.43	0.26	0.51	0.46	0.73	0.23	0.47	0.60
R	0.32	0.20	0.18	0.17	0.38	0.24	0.46	0.48	0.52	0.20	0.51	0.58
FID	342	36	1101	1373	1511	112	1539	1848	1495	120	629	795
Precission	0.98	1.00	0.99	0.99	0.79	0.99	0.99	0.98	0.56	0.96	0.99	0.98
Recall	0.98	1.00	0.90	0.88	0.98	1.00	0.78	0.77	0.97	1.00	0.75	0.72

Table 12: ImageNet subset tests for three feature-layer of the VGG-16 network given 25000 ImageNet images.

	Layer 2				Layer 3				Layer 4			
Truncation	0	0.3	0.7	1	0	0.3	0.7	1	0	0.3	0.7	1
I	-8.67	-3.46	-1.00	0.66	-3.37	0.07	-2.87	0.67	-16.24	0.18	0.07	0.65
G	0.75	0.38	0.23	0.11	0.82	0.59	0.49	0.19	0.92	0.83	0.54	0.18
R	0.93	0.54	0.35	0.21	0.89	0.40	0.25	0.19	0.91	0.47	0.26	0.13
FID	74xe4	33xe4	49xe3	1746.8	12xe4	57xe3	9485.8	326.30	15xe3	6498.8	925.89	61.53
Precission	0.04	0.47	0.92	0.99	0.02	0.44	0.91	0.99	0.00	0.40	0.94	0.99
Recall	0.33	0.72	0.97	0.99	0.12	0.79	0.97	1.00	0.10	0.70	0.97	1.00

Table 13: Truncation tests for three feature-layer of the InceptionNet-V3 given 50000 FFHQ images.

	Layer 2				Layer 3				Layer 4			
Truncation	0	0.3	0.7	1	0	0.3	0.7	1	0	0.3	0.7	1
I	-2.48	0.11	0.28	0.61	-0.22	0.31	0.47	0.75	-0.06	0.28	0.52	0.59
G	0.72	0.85	0.65	0.44	0.74	0.68	0.63	0.23	0.79	0.79	0.63	0.39
R	0.97	0.67	0.46	0.35	0.87	0.54	0.11	0.18	0.82	0.47	0.10	0.33
FID	14xe6	74xe5	18xe5	31xe4	40xe5	20xe5	53xe4	96xe3	16xe4	95xe3	27xe3	50xe2
Precission	0.02	0.35	0.86	0.98	0.01	0.37	0.86	0.97	0.00	0.36	0.87	0.97
Recall	0.37	0.75	0.97	1.00	0.13	0.63	0.97	0.99	0.09	0.61	0.95	0.99

Table 14: Truncation tests for three feature-layer of the VGG-16 network given 50000 FFHQ images.

	Layer 2				Layer 3				Layer 4			
Truncation	0	0.3	0.7	1	0	0.3	0.7	1	0	0.3	0.7	1
I	-10.79	-5.57	-0.16	0.33	-52.78	-3.28	-0.95	-0.15	-14.49	-14.84	-1.53	-0.62
G	0.60	0.35	0.42	0.47	0.70	0.59	0.62	0.58	0.91	0.85	0.64	0.62
R	0.89	0.70	0.20	0.10	0.89	0.76	0.36	0.34	0.94	0.83	0.59	0.52
FID	10357	6347.7	2786.0	1637.3	3871.4	1972.5	606.37	354.74	1619.4	859.30	295.41	185.38
Precission	0.12	0.47	0.87	0.95	0.03	0.38	0.88	0.96	0.00	0.10	0.75	0.92
Recall	0.65	0.82	0.88	0.86	0.19	0.57	0.89	0.86	0.01	0.19	0.85	0.89

Table 15: Truncation tests for three feature-layer of the InceptionNet-V3 given 38676 ultra-violet solar images.

	Layer 2				Layer 3				Layer 4			
Truncation	0	0.3	0.7	1	0	0.3	0.7	1	0	0.3	0.7	1
I	0.00	-12.64	-4.52	-0.11	-9.36	-3.99	-0.14	0.18	-0.63	-0.27	-0.11	0.12
G	1.00	0.29	0.22	0.49	0.64	0.45	0.50	0.52	0.84	0.72	0.58	0.62
R	1.00	0.91	0.82	0.61	0.82	0.59	0.47	0.41	0.78	0.65	0.56	0.53
FID	27xe4	16xe4	93xe3	95744	76142	44480	23056	24017	2424.9	1505.6	732.15	711.57
Precission	0.03	0.47	0.73	0.77	0.02	0.37	0.79	0.85	0.02	0.38	0.87	0.93
Recall	0.33	0.76	0.92	0.86	0.28	0.76	0.91	0.81	0.42	0.69	0.80	0.78

Table 16: Truncation tests for three feature-layer of the VGG-16 network given 38676 ultra-violet solar images.